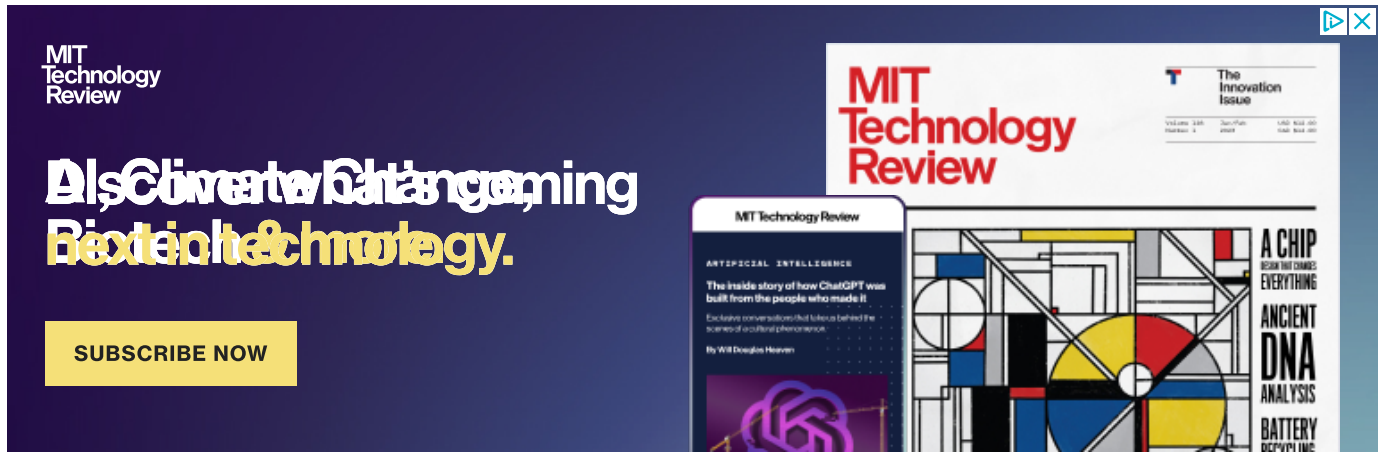




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Large language models can do jaw-dropping things. But nobody knows exactly why.

And that's a problem. Figuring it out is one of the biggest scientific puzzles of our time and a crucial step towards controlling more powerful future models.

By Will Douglas Heaven

March 4, 2024



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Two years ago, Yuri Burda and Harri Edwards, researchers at the San Francisco–based firm OpenAI, were trying to find out what it would take to get a language model to do basic arithmetic. They wanted to know how many examples of adding up two numbers the model needed to see before it was able to add up any two numbers they gave it. At first, things didn't go too well. The models memorized the sums they saw but failed to solve new ones.

By accident, Burda and Edwards left some of their experiments running far longer than they meant to—days rather than hours. The models were shown the example sums over and over again, way past the point when the researchers would otherwise have called it quits. But when the pair at last came back, they were surprised to find that the experiments had worked. They'd trained a language model to add two numbers—it had just taken a lot more time than anybody thought it should.

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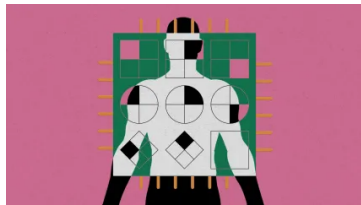
Curious about what was going on, Burda and Edwards teamed up with colleagues to study the phenomenon. They found that in certain cases, models could seemingly fail to learn a task and then all of a sudden just get it, as if a lightbulb had switched on. This wasn't how deep learning was supposed to work. They called the behavior grokking.

"It's really interesting," says Hattie Zhou, an AI researcher at the University of Montreal and Apple Machine Learning Research, who wasn't involved in the work. "Can we ever be confident that models have stopped learning? Because maybe we just haven't trained for long enough."

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The weird behavior has captured the imagination of the wider research community. "Lots of people have opinions," says Lauro Langosco at the University of Cambridge, UK. "But I don't think there's a consensus about what exactly is going on."

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Grokking is just one of several odd phenomena that have AI researchers scratching their heads. The largest models, and large language models in particular, seem to behave in ways textbook math says they shouldn't. This highlights a remarkable fact about deep learning, the fundamental technology behind today's AI boom: for all its runaway success, nobody knows exactly how—or why—it works.

"Obviously, we're not completely ignorant," says Mikhail Belkin, a computer scientist at the University of California, San Diego. "But our theoretical analysis is so far off what these models can do. Like, why can they learn language? I think this is very mysterious."

The biggest models are now so complex that researchers are studying them as if they were strange natural phenomena, carrying out experiments and trying to explain the results. Many of those observations fly in the face of classical statistics, which had provided our best set of explanations for how predictive models behave.

So what, you might say. In the last few weeks, Google DeepMind has rolled out its generative models across most of its consumer apps. OpenAI wowed people with Sora, its stunning new text-to-video model. And businesses around the world are scrambling to co-opt AI for their needs. The tech works— isn't that enough?

But figuring out why deep learning works so well isn't just an intriguing scientific puzzle. It could also be key to unlocking the next generation of the technology—as well as getting a handle on its formidable risks.

"These are exciting times," says Boaz Barak, a computer scientist at Harvard University who is on secondment to OpenAI's superalignment team for a year. "Many people in the field often compare it to physics at the beginning of the 20th century. We have a lot of experimental results that we don't completely understand, and often when you do an experiment it surprises you."

Old code, new tricks

Most of the surprises concern the way models can learn to do things that they have not been shown how to do.

Known as generalization, this is one of the most fundamental ideas in machine learning—and its greatest puzzle. Models learn to do a task—spot faces, translate sentences, avoid pedestrians—by training with a specific set of examples. Yet they can generalize, learning to do that task with examples they have not seen before. Somehow, models do not just memorize patterns they have seen but come up with rules that let them apply those patterns to new cases. And sometimes, as with grokking, generalization happens when we don't expect it to.

Large language models in particular, such as OpenAI's GPT-4 and Google DeepMind's Gemini, have an astonishing ability to generalize. "The magic is not that the model can learn math problems in English and then generalize to new math problems in English," says Barak, "but that the model can learn math problems in English, then see some French literature, and from that generalize to solving math problems in French. That's something beyond what statistics can tell you about."

When Zhou started studying AI a few years ago, she was struck by the way her teachers focused on the how but not the why. "It was like, here is how you train these models and then here's the result," she says. "But it wasn't clear why this process leads to models that are capable of doing these amazing things." She wanted to know more, but she was told there weren't good answers: "My assumption was that scientists know what they're doing. Like, they'd get the theories and then they'd build the models. That wasn't the case at all."



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The rapid advances in deep learning over the last 10-plus years came more from trial and error than from understanding. Researchers copied what worked for others and tacked on innovations of their own. There are now many different ingredients that can be added to models and a growing cookbook filled with recipes for using them. "People try this thing, that thing, all these tricks," says Belkin. "Some are important. Some are probably not."

"It works, which is amazing. Our minds are blown by how powerful these things are," he says. And yet for all their success, the recipes are more alchemy than chemistry: "We figured out certain incantations at midnight after mixing up some ingredients," he says.

Overfitting

The problem is that AI in the era of large language models appears to defy textbook statistics. The most powerful models today are vast, with up to a trillion parameters (the values in a model that get adjusted during training). But statistics says that as models get bigger, they should first improve in performance but then get worse. This is because of something called overfitting.

When a model gets trained on a data set, it tries to fit that data to a pattern. Picture a bunch of data points plotted on a chart. A pattern that fits the data can be represented on that chart as a line running through the points. The process of training a model can be thought of as getting it to find a line that fits the training data (the dots already on the chart) but also fits new data (new dots).

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A straight line is one pattern, but it probably won't be too accurate, missing some of the dots. A wiggly line that connects every dot will get full marks on the training data, but won't generalize. When that happens, a model is said to overfit its data.

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According to classical statistics, the bigger a model gets, the more prone it is to overfitting. That's because with more parameters to play with, it's easier for a model to hit on wiggly lines that connect every dot. This suggests there's a sweet spot between under- and overfitting that a model must find if it is to generalize. And yet that's not what we see with big models. The best-known example of this is a phenomenon known as double descent.

The performance of a model is often represented in terms of the number of errors it makes: as performance goes up, error rate goes down (or descends). For decades, it was believed that error rate went down and then up as models got bigger: picture a U-shaped curve with the sweet spot for generalization at the lowest point. But in 2018, Belkin and his colleagues found that when certain models got bigger, their error rate went down, then up—and then down again (a double descent, or W-shaped curve). In other words, large models would somehow overrun that sweet spot and push through the overfitting problem, getting even better as they got bigger.

A year later, Barak coauthored a paper showing that the double-descent phenomenon was more common than many thought. It happens not just when models get bigger but also in models with large amounts of training data or models that are trained for longer. This behavior, dubbed benign overfitting, is still not fully understood. It raises basic questions about how models should be trained to get the most out of them.

Researchers have sketched out versions of what they think is going on. Belkin believes there's a kind of Occam's razor effect in play: the simplest pattern that fits the data—the smoothest curve between the dots—is often the one that generalizes best. The reason bigger models keep improving longer than it seems they should could be that bigger models are more likely to hit upon that just-so curve than smaller ones: more parameters means more possible curves to try out after ditching the wiggliest.

“Our theory seemed to explain the basics of why it worked,” says Belkin. “And then people made models that could speak 100 languages and it was like, okay, we understand nothing at all.” He laughs: “It turned out we weren't even scratching the surface.”



How China Got Ahead on EVs

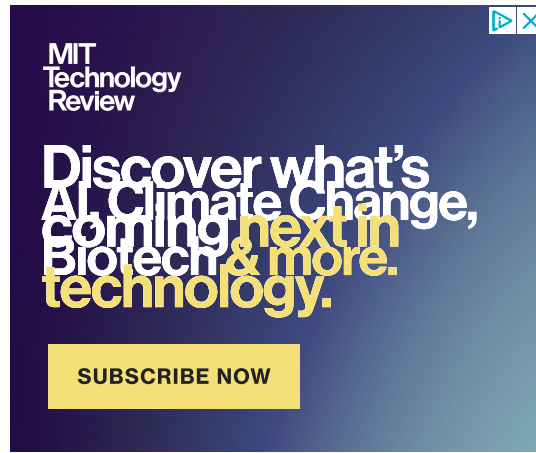
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For Belkin, large language models are a whole new mystery. These models are based on transformers, a type of neural network that is good at processing sequences of data, like words in sentences.

There's a lot of complexity inside transformers, says Belkin. But he thinks at heart they do more or less the same thing as a much better understood statistical construct called a Markov chain, which predicts the next item in a sequence based on what's come before. But that isn't enough to explain everything that large language models can do. “This is something that, until recently, we thought should not work,” says Belkin. “That means that something was fundamentally missing. It identifies a gap in our understanding of the world.”

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Belkin goes further. He thinks there could be a hidden mathematical pattern in language that large language models somehow come to exploit: “Pure speculation but why not?”

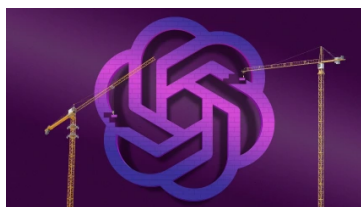
“The fact that these things model language is probably one of the biggest discoveries in history,” he says. “That you can learn language by just predicting the next word with a Markov chain—that’s just shocking to me.”

Start small

Researchers are trying to figure it out piece by piece. Because large models are too complex to study themselves, Belkin, Barak, Zhou, and others experiment instead on smaller (and older) varieties of statistical model that are better understood. Training these proxies under different conditions and on various kinds of data and observing what happens can give insight into what’s going on. This helps get new theories off the ground, but it is not always clear if those theories will hold for larger models too. After all, it is in the complexity of large models that many of the weird behaviors reside.

Is a theory of deep learning coming? Daniel Hsu, a computer scientist at Columbia University who was one of Belkin’s coauthors on the double-descent paper, doesn’t expect all the answers anytime soon. “We have better intuition now,” he says. “But really explaining everything about why neural networks have this kind of unexpected behavior? We’re still far from doing that.”

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In 2016, Chiyuan Zhang at MIT and colleagues at Google Brain published an [influential paper](#) titled “Understanding Deep Learning Requires Rethinking Generalization.” In 2021, five years later, they republished the paper, calling it “Understanding Deep Learning (Still) Requires Rethinking Generalization.” What about in 2024? “Kind of yes and no,” says Zhang. “There has been a lot of progress lately, though probably many more questions arise than get resolved.”

Meanwhile, researchers continue to wrestle even with the basic observations. In December, Langosco and his colleagues presented a paper at NeurIPS, a top AI conference, in which they claimed that grokking and double descent are in fact aspects of the same phenomenon. “You eyeball them and they look kind of similar,” says Langosco. He believes that an explanation of what’s going on should account for both.

At the same conference, Alicia Curth, who studies statistics at the University of Cambridge, and her colleagues argued that double descent is in fact an illusion. “It didn’t sit very well with me that modern machine learning is some kind of magic that defies all the laws that we’ve established so far,” says Curth. Her team argued that the double-descent phenomenon—where models appear to perform better, then worse, and then better again as they get bigger—arises because of the way the complexity of the models was measured.

Belkin and his colleagues used model size—the number of parameters—as a measure of complexity. But Curth and her colleagues found that the number of parameters might not be a good stand-in for complexity because adding parameters sometimes makes a model more complex and sometimes makes it less so. It depends what the values are, how they get used during training, and how they interact with others—much of which stays hidden inside the model. “Our takeaway was that not all model parameters are created equal,” says Curth.

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In short, if you use a different measure for complexity, large models might conform to classical statistics just fine. That’s not to say there isn’t a lot we don’t understand about what happens when models get bigger, says Curth. But we already have all the math we need to explain it.

A great mystery of our time

It’s true that such debates can get into the weeds. Why does it matter whether AI models are underpinned by classical statistics or not?

One answer is that better theoretical understanding would help build even better AI or make it more efficient. At the moment, progress has been fast but unpredictable. Many things that OpenAI’s GPT-4 can do came as a surprise even to the people who made it. Researchers are still arguing over what it can and cannot achieve. “Without some sort of fundamental theory, it’s very hard to have any idea what we can expect from these things,” says Belkin.

Barak agrees. “Even once we have the models, it is not straightforward even in hindsight to say exactly why certain capabilities emerged when they did,” he says.

This isn’t only about managing progress—it’s about anticipating risk, too. Many of the researchers working on the theory behind deep learning are motivated by safety concerns for future models. “We don’t know what capabilities GPT-5 will have until we train it and test it,” says Langosco. “It might be a medium-size problem right now, but it will become a really big problem in the future as models become more powerful.”

Barak works on OpenAI’s superalignment team, which was set up by the firm’s chief scientist, Ilya Sutskever, to figure

out how to stop a hypothetical superintelligence from going rogue. “I’m very interested in getting guarantees,” he says. “If you can do amazing things but you can’t really control it, then it’s not so amazing. What good is a car that can drive 300 miles per hour if it has a shaky steering wheel?”

But beneath all that there’s also a grand scientific challenge. “Intelligence is definitely up there as one of the great mysteries of our time,” says Barak.

“We’re a very infant science,” he says. “The questions that I’m most excited about this month might be different to the questions that I’m most excited about next month. We are still discovering things. We very much need to experiment and get surprised.” **T**

by Will Douglas Heaven



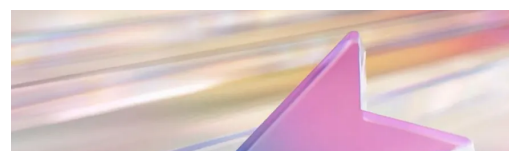
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